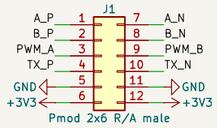
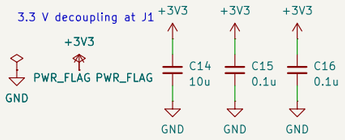
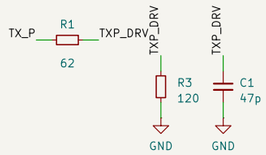
Pmod 100BASE-TX PHY front end, MagJack (HR911105A) variant — see ../README.md

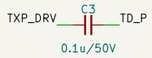
Pmod J1: plugs into Tang Primer 25K pmod0 (3.3 V bank). 1-4 top row = P, 7-10 bottom row = N.
A: 1/7 (F5/G5) B: 2/8 (G7/G8) PWM_A/B: 3/9 (H8/H7) TX_P/N: 4/10 (H5/J5) — verify by ball name.

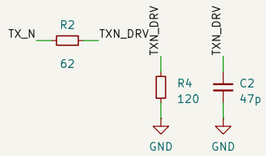


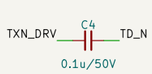


TX: LVCMOS33 x2 -> Rs 62R / Rp 120R (Thevenin 2V/50R each side) -> Cr 47pF -> C3/C4 0.1uF DC block -> transformer primary (J2 1/2). The DC block is REQUIRED: +/- states put DC across the winding.

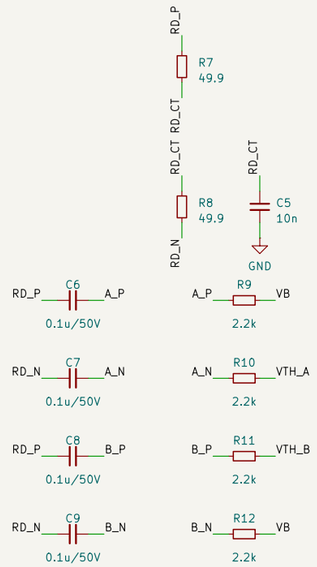




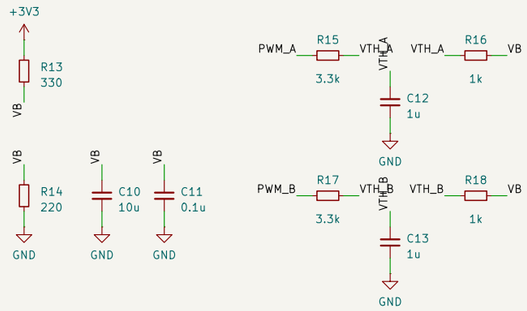




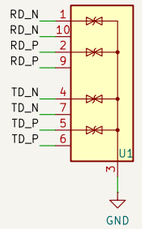
RX: J2 3/6 (transformer secondary) -> 2x49.9R termination, midpoint Ct to GND.
Tap caps Ca 0.1uF make 4 DC-isolated comparator inputs.
A = (v > VTH_A-VB), B = (v > -(VTH_B-VB)), c = A^B = NRZI.



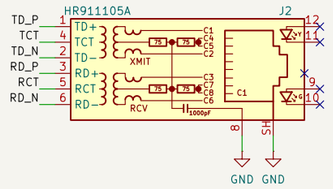
Bias: VB = 1.32 V. VTH = VB + (D*3.3-VB)*1k/4.3k (-0.31 .. +0.46 V vs VB)

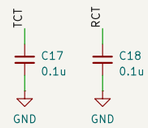


U1: ESD array on the PHY side of the magnetics (flow-through, NC pads carry the line).
TCT/RCT: AC ground only (C17/C18). Do NOT tie to 3.3 V (voltage-mode driver).
J2 pin 8 = integrated Bob Smith 1nF/2kV return -> GND. LEDs (9-12) unused.
PoE: no PSE behind this board; isolation by magnetics.



TPD4E02B04





Peer must have magnetics (normal switch / NIC). Not for board-to-board direct use.
Transformerless (capacitor coupled) 100BASE-TX front end for Tang Primer 25K

Kenta IDA

Sheet: /
File: pmod_100base_tx_magjack.kicad_sch

Title: Pmod 100BASE-TX direct-attach PHY

Size: A3	Date: 2026-10-10	**Rev: Rev.A**
KiCad E.D.A. 10.0.6		Id: 1/1